

Global Sourcing Model for AI Microchips

Strategic Supplier Region Analysis for a U.S.-Based AI Chip Company

Prepared for portfolio presentation

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1. Executive Summary

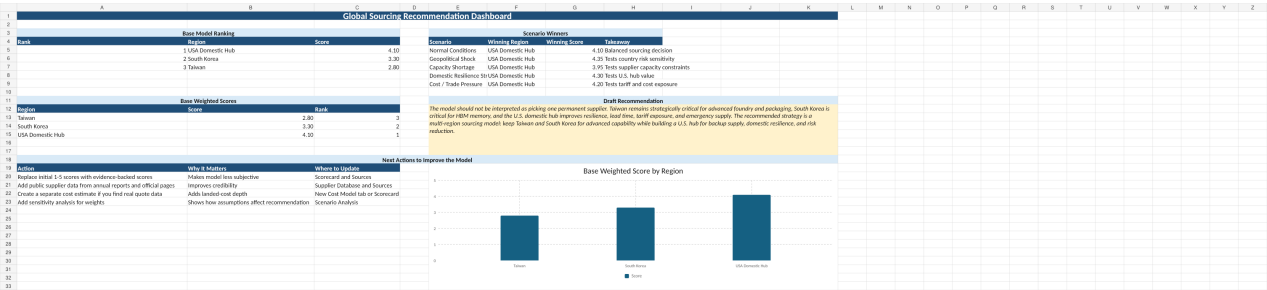
This report explains and supports the global sourcing workbook built for an Nvidia-style U.S. AI chip company. The model compares Taiwan, South Korea, and a U.S. Domestic Hub for sourcing an AI microchip / GPU supply chain. It evaluates each region across geopolitical risk, lead time risk, tariff and trade cost, contract lead time, capacity availability, supplier criticality, and substitution difficulty.

The model does not recommend replacing Taiwan or South Korea with the United States. That would be an unrealistic conclusion because Taiwan remains critical for advanced foundry and packaging capability, while South Korea remains critical for high-bandwidth memory and semiconductor manufacturing. Instead, the correct recommendation is a multi-region sourcing strategy: keep Taiwan and South Korea for advanced capability while building a U.S. Domestic Hub for resilience, risk reduction, faster replenishment, and emergency supply support.

The base scorecard ranks USA Domestic Hub first at 4.10, South Korea second at 3.30, and Taiwan third at 2.80. This ranking reflects the model design: the U.S. option scores well on geopolitical exposure, lead time, tariff exposure, and substitution flexibility, while Taiwan and South Korea score higher in supplier criticality and advanced capability.

Rank	Region	Base Weighted Score	Interpretation
1	USA Domestic Hub	4.10	Best resilience option, not full capability replacement
2	South Korea	3.30	Critical HBM and semiconductor supplier region
3	Taiwan	2.80	Critical foundry and packaging region with higher concentration risk

Dashboard preview from workbook



2. Project Context and Purpose

The purpose of this case study is to show how global sourcing decisions are made when the product is high-value, capacity constrained, technologically advanced, and exposed to geopolitical and trade risk. The product being sourced is an AI microchip / GPU supply chain, not a simple microcontroller or retail electronic component.

The case company is modeled as a U.S.-based Nvidia-style AI chip company. This matters because the company depends on advanced semiconductor manufacturing, advanced packaging, memory suppliers, wafers, assembly partners, and domestic distributors for different parts of the supply chain. A strong sourcing strategy must understand the role of each supplier layer instead of treating all sellers as interchangeable.

Field	Input / Assumption	Notes
Case Company	Nvidia-style U.S. AI chip company	Use Nvidia as the case company for strategic global sourcing learning.
Buyer Location	United States	The buyer is based in the U.S.
Product Being Sourced	AI microchip / GPU supply chain	Advanced microchip supply chain, not a simple retail MCU.
Supplier A Region	Taiwan	Foundry, advanced packaging, wafers, server assembly exposure.
Supplier B Region	South Korea	HBM memory, semiconductor manufacturing, wafers.
Supplier C Region	USA Domestic Hub	Domestic manufacturing, local sellers, backup supply, resilience.
Score Scale	1 to 5	5 = best or lowest risk. 1 = worst or highest risk.
Base Time Horizon	12 to 24 months	Use this to frame risk and lead time assumptions.
Currency	USD	Model is strategic, so most costs are qualitative unless you add real prices.
Main Limitation	Supplier contracts and exact prices are not public	Use public sources, company reports, and reasoned assumptions.

3. Methodology and Scoring Logic

The model uses a weighted scorecard. Each region receives a 1 to 5 score in each category, where 5 means the strongest performance or lowest risk and 1 means the weakest performance or highest risk. The final weighted score is calculated by multiplying each category score by its assigned weight, then summing the weighted values.

Base formula logic:

Weighted Score = SUM(Category Score x Category Weight)

Category	Weight	Why It Matters
Geopolitical Risk	20%	Measures exposure to regional conflict, policy restrictions, and cross-border disruption.
Lead Time Risk	15%	Measures how reliably the supplier region can support planning and delivery timelines.
Tariff / Trade Cost	10%	Measures import cost exposure, trade restrictions, and landed-cost uncertainty.
Contract Lead Time	10%	Measures how quickly the buyer can access, onboard, or negotiate capacity.
Capacity Availability	20%	Measures whether available production capacity can support AI chip demand.
Supplier Criticality	15%	Measures how essential the region is to the AI chip supply chain.
Substitution Difficulty	10%	Measures how difficult it would be to replace the supplier region in the short term.

The weights show that the model values resilience and capacity heavily. Geopolitical risk and capacity availability each receive 20 percent because a disruption in either area can prevent product delivery even when demand exists. Supplier criticality receives 15 percent because certain suppliers cannot be easily replaced. Lead time receives 15 percent because long or unreliable lead times create planning risk. Tariff, contract, and substitution factors receive smaller but still meaningful weights.

4. Supplier Region Analysis

The supplier database separates the supply chain into direct strategic suppliers, upstream suppliers, downstream assembly partners, domestic hubs, and local sellers. This is important because not every company in the database

plays the same role. TSMC, SK hynix, and Samsung are strategic semiconductor suppliers, while DigiKey and Mouser are useful domestic distributors but not realistic sources for advanced Nvidia-class AI GPUs.

Taiwan

Taiwan is treated as a critical capability region because of TSMC, advanced packaging, wafer exposure, and system assembly partners. Its lower score is not because the region lacks capability. It is because the model penalizes concentration risk, geopolitical exposure, capacity bottlenecks, and short-term substitution difficulty. Taiwan scores high in supplier criticality but lower in geopolitical risk and substitution flexibility.

South Korea

South Korea is treated as a critical memory and semiconductor region. SK hynix and Samsung make the region highly important for HBM memory and semiconductor diversification. Its model score sits between Taiwan and the U.S. Domestic Hub because it has strong capability but still carries overseas lead time, trade, and capacity exposure.

USA Domestic Hub

The USA Domestic Hub is treated as a resilience layer, not as a complete replacement for Asian semiconductor capability. It scores highly on geopolitical exposure, lead time, tariff exposure, and substitution flexibility because domestic or regional capacity can reduce disruption risk. However, it receives lower criticality and capacity scores than Taiwan/South Korea because the U.S. hub is still developing and may not fully replace advanced offshore suppliers in the short term.

Region	Company	Role	Product / Service	Layer	Criticality Rationale
Taiwan	TSMC	Foundry manufacturing	Advanced GPU / AI chip manufacturing	Direct strategic supplier	Advanced foundry capability is difficult to substitute.
Taiwan	TSMC CoWoS	Advanced packaging	CoWoS packaging for AI chips	Direct strategic supplier	Advanced packaging can become a capacity bottleneck.
Taiwan	GlobalWafers	Silicon wafers	Silicon wafers	Upstream supplier	Wafer supply affects semiconductor output and lead times.
Taiwan	Foxconn	System assembly	AI servers and electronics manufacturing	Downstream manufacturing partner	Assembly capacity affects delivery of final systems.
Taiwan	Wistron	System assembly	Servers and electronics manufacturing	Downstream manufacturing partner	Potential assembly partner for AI infrastructure.
South Korea	SK hynix	HBM memory	High-bandwidth memory	Direct strategic supplier	HBM is critical for AI accelerators.
South Korea	Samsung Electronics	Memory / foundry	HBM, DRAM, foundry services	Direct or alternate supplier	Provides memory and semiconductor capacity.
South Korea	SK Siltron	Silicon wafers	Silicon wafers	Upstream supplier	Wafer supplier supporting semiconductor manufacturing.
USA	TSMC Arizona	Domestic foundry hub	U.S.-made advanced wafers / chips	Domestic hub	Reduces cross-border exposure for some production.
USA	Micron	Memory	DRAM / HBM roadmap exposure	Domestic / allied memory option	Potential U.S.-based memory diversification.
USA	Amkor	Advanced packaging	Semiconductor packaging and test	Domestic / regional hub	Packaging can reduce reliance on overseas bottlenecks.
USA	Intel Foundry	Foundry manufacturing	Domestic foundry services	Domestic alternative	Potential long-term domestic foundry alternative.
USA	DigiKey	Local distributor	Electronic components	Local seller	Useful for lower-level components and development needs.
USA	Mouser	Local distributor	Electronic components	Local seller	Useful for small-volume component sourcing.
USA	Arrow Electronics	Distributor / supply chain services	Electronic components and services	Local seller	Useful for sourcing support and supply chain services.
USA	Avnet	Distributor / supply chain services	Electronic components and services	Local seller	Useful for electronics sourcing and supply chain services.

5. Base Model Scorecard

The base model ranking is useful, but it should be interpreted carefully. A higher score does not automatically mean the region is the only supplier choice. It means the region performs better under the selected scorecard weights. In this model, the U.S. Domestic Hub wins because the base weights value resilience, lead time, tariff exposure, and risk reduction.

Region	Geo	Lead	Trade	Contract	Capacity	Criticality	Substitution	Score	Rank
Taiwan	2	3	4	3	2	5	1	2.80	3
South Korea	3	3	4	3	3	5	2	3.30	2
USA Domestic Hub	5	5	5	4	3	3	4	4.10	1

The Taiwan result is the best example of why the model needs explanation. Taiwan receives a high criticality score, but the weighted score is pulled down by geopolitical risk, capacity constraints, and substitution difficulty. This means Taiwan is strategically necessary but risky as a concentrated dependency.

The U.S. Domestic Hub result also needs explanation. It should not be read as proof that U.S. suppliers can replace TSMC, SK hynix, or Samsung. It means a domestic hub improves resilience and may be valuable as a risk-control strategy, emergency sourcing layer, and partial manufacturing or distribution backup.

Scorecard preview from workbook

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	Base Supplier Region Scorecard																									
2	Category		Weight																							
3	Geopolitical Risk		20%																							
4	Lead Time Risk		15%																							
5	Tariff / Trade Cost		10%																							
6	Contract Lead Time		10%																							
7	Capacity Availability		20%																							
8	Supplier Criticality		15%																							
9	Substitution Difficulty		10%																							
10																										
11																										
12	Supplier Region	Main Commodities	Main Role	Geo Risk	Lead Time	Tariff / Trade	Contract Lead	Capacity	Criticality	Substitution	Weighted Score	Rank														
13	Taiwan	TSMC, GlobalFoundries, Foxconn, Wistron	Foundry, packaging, wafers, assembly	2	3	4	3	2	5	1	2.80	3														
14	South Korea	SK hynix, Samsung, SK Siltron	HBM memory, memory/foundry, wafers	3	3	4	3	3	5	2	3.30	2														
15	USA Domestic Hub	TSMC Arizona, Micron, Intel, Intel	Domestic hub, backup, local wafers	5	5	5	4	3	3	4	4.10	1														

6. Scenario Analysis

The scenario analysis tests whether the recommendation changes when the weight structure changes. This is important because global sourcing decisions depend on the business environment. A company facing normal conditions may weigh factors differently than a company facing geopolitical shock, capacity shortage, tariff pressure, or a domestic resilience strategy.

Scenario	Geo Wt	Lead Wt	Tariff Wt	Contract Wt	Capacity Wt	Criticality Wt	Substitution Wt	Check
Normal Conditions	20%	15%	10%	10%	20%	15%	10%	1
Geopolitical Shock	35%	15%	10%	5%	15%	10%	10%	1
Capacity Shortage	15%	20%	5%	10%	35%	10%	5%	1
Domestic Resilience Strategy	25%	20%	10%	10%	15%	10%	10%	1
Cost / Trade Pressure	15%	10%	25%	10%	20%	10%	10%	1
Scenario	Winning Region	Winning Score	Lowest Region		Lowest Score	Logic		
Normal Conditions	USA Domestic Hub	4.10	Taiwan		2.80	Resilience weights favor U.S. hub.		
Geopolitical Shock	USA Domestic Hub	4.35	Taiwan		2.60	Resilience weights favor U.S. hub.		
Capacity Shortage	USA Domestic Hub	3.95	Taiwan		2.65	Resilience weights favor U.S. hub.		
Domestic Resilience Strategy	USA Domestic Hub	4.30	Taiwan		2.70	Resilience weights favor U.S. hub.		
Cost / Trade Pressure	USA Domestic Hub	4.20	Taiwan		2.90	Resilience weights favor U.S. hub.		

The most important insight is that the U.S. Domestic Hub wins across every modeled scenario. That does not prove the U.S. should become the only sourcing region. It proves the model is currently weighted toward resilience and risk control. A more capability-heavy model could produce a different conclusion, especially if advanced foundry capability and HBM availability were weighted more heavily.

7. Local Seller and Domestic Distributor Role

The local seller section strengthens the project because it shows that domestic supply chains include distributors and service providers, not only chip manufacturers. However, this section must be framed correctly. DigiKey, Mouser, Arrow, and Avnet are not normal sources for advanced Nvidia-class AI GPUs. Their best use is domestic component sourcing, engineering samples, prototype support, replacement parts, and broader electronics supply chain services.

Seller	Country	Role	Best Use Case	Limitations	Authorized / Reliability Score	Lead Time Score	Notes
DigiKey	USA	Electronic component distributor	Small-volume components, dev boards, replacement parts	Not a normal source for advanced Nvidia AI GPUs	5	5	Good for learning domestic sourcing and inventory availability.
Mouser	USA	Electronic component distributor	Electronic components and engineering samples	Not a normal source for advanced Nvidia AI GPUs	5	5	Good for comparing price breaks and availability.
Arrow Electronics	USA	Distributor and supply chain services	Larger component sourcing and supply chain programs	Pricing and contracts may be quote-based	4	4	Useful for sourcing support and business-to-business procurement.
Avnet	USA	Distributor and supply chain services	Electronics sourcing and supply chain services	Pricing and contracts may be quote-based	4	4	Useful for strategic sourcing support.

8. Risk Matrix and Mitigation Logic

The risk matrix converts the scorecard into operational thinking. A scorecard shows which region ranks highest, but the risk matrix explains what could go wrong and what the sourcing team should do about it. This is one of the strongest parts of the workbook because it shows decision support rather than simple ranking.

Risk	Taiwan Exposure	South Korea Exposure	USA Domestic Hub Exposure	Impact	Mitigation
Geopolitical disruption	High	Medium	Low	Could disrupt foundry, packaging, and delivery timelines	Dual-source by region; increase domestic capacity; monitor Taiwan Strait risk
Capacity shortage	High	Medium-high	Medium	Could limit AI chip output even if demand exists	Long-term capacity agreements; reserve capacity; qualify alternates
HBM memory bottleneck	Low-medium	High	Medium	Could delay AI chip production and increase costs	Use multiple memory suppliers; increase inventory buffer; monitor HBM roadmap
Advanced packaging bottleneck	High	Medium	Medium	Could delay completed AI accelerators	Develop alternate packaging partners; use U.S. packaging where possible
Tariff / trade policy change	Medium	Medium	Low	Could raise landed cost or restrict imports	HTS review; trade compliance monitoring; domestic hub option
Lead time delay	Medium-high	Medium	Low	Could delay customer shipments	Safety stock; split sourcing; faster domestic replenishment
Supplier concentration	High	Medium	Low-medium	Creates single point of failure	Qualify second sources; build supplier scorecard and trigger points
Contract lead time	Medium	Medium	Low	Delays onboarding or capacity access	Framework agreements; pre-approved supplier terms

9. Final Recommendation

The recommended sourcing strategy is not to choose one permanent supplier region. The correct recommendation is a multi-region sourcing model with three roles: Taiwan for advanced foundry and packaging, South Korea for HBM memory and semiconductor diversification, and the U.S. Domestic Hub for resilience, backup capacity, local distribution, and risk reduction.

- Keep Taiwan as a core strategic supplier region for advanced foundry, advanced packaging, wafers, and assembly exposure.
- Keep South Korea as a core strategic supplier region for HBM memory, memory/foundry optionality, and wafer supply.

3. Build the U.S. Domestic Hub as a resilience layer rather than a full replacement. Use it for domestic capacity, emergency sourcing, faster replenishment, distributor support, and reduced trade exposure.
4. Use long-term capacity agreements, dual-region sourcing, inventory buffers, and supplier qualification triggers to reduce dependency on any single region.
5. Upgrade the model over time by replacing subjective scores with evidence-backed scores from annual reports, official supplier pages, trade data, and real lead-time or quote information.

The strongest portfolio wording would be: Strategic global sourcing model comparing Taiwan, South Korea, and a U.S. Domestic Hub for AI microchip supply chain resilience. This phrasing is more accurate than saying the project simply chooses the United States as the best supplier.

10. Portfolio Readiness Review

This project is portfolio-worthy because it demonstrates a real supply chain decision framework. It has assumptions, supplier mapping, a weighted scorecard, scenario analysis, local distributor comparison, risk mitigation, dashboard summaries, and source tracking. That is much stronger than a basic spreadsheet because it shows how sourcing decisions are structured and defended.

Current readiness rating: 8/10. The project is strong enough to include in a portfolio, but it should be presented as an analytical sourcing model rather than a perfect real-world procurement answer.

Strength		Why It Helps the Portfolio
Clear sourcing problem		The project has a specific product, buyer location, and supplier regions.
Weighted decision model		Shows structured evaluation instead of opinion-based ranking.
Scenario analysis		Shows how recommendations change when business conditions change.
Supplier database		Connects sourcing strategy to real companies and supply chain layers.
Risk matrix		Turns the recommendation into action and mitigation planning.
Dashboard		Makes the model easier to explain visually.
Weakness to Fix	Why It Matters	How to Improve
Some scores are still subjective	Recruiters may question whether the rankings are evidence-based.	Add a score rationale column with source-backed reasoning for every score.
U.S. hub can be misread as full replacement	This would be unrealistic for advanced AI chips.	State clearly that the U.S. hub is a resilience layer, not a complete substitute.
Local sellers may be misunderstood	DigiKey and Mouser are not AI GPU suppliers.	Position them as component distributors and support channels.
Dashboard still looks spreadsheet-like	Portfolio projects should be visually clean and executive-ready.	Add cleaner charts, spacing, and recommendation boxes.
No real cost model yet	Tariff and landed-cost logic would be stronger with numbers.	Add a landed-cost or quote-estimate tab if public data becomes available.

11. Limitations and Next Improvements

The model is strong as a strategic learning project, but it has limitations. Exact AI chip supplier contracts, capacity allocations, delivery lead times, and pricing are usually not public. Because of that, the current workbook uses public sources, reasoned assumptions, and qualitative scoring. This is acceptable for a portfolio case study as long as the report clearly explains the assumption-based nature of the model.

- Add a Scoring Evidence tab with one short explanation and source for every 1 to 5 score.
- Add sensitivity analysis showing how the ranking changes if capability or criticality receives more weight.
- Add a cost model if real quote data, tariff information, or proxy cost estimates can be found.
- Add a supplier risk heatmap that plots impact against probability.
- Add a final executive dashboard with one recommendation box, one score chart, one scenario winner table, and one risk summary.

- Separate true AI chip supply from support-level electronics distribution so the model is not overclaiming what domestic sellers can supply.

Appendix A: Workbook Data Tables

A1. Scoring Guide

Category	5 = Best / Lowest Risk	4 = Good	3 = Moderate	2 = High Risk	1 = Very High Risk
Geopolitical Risk	Domestic / stable policy environment	Low cross-border risk	Some regional or policy risk	High regional exposure	Severe geopolitical concentration
Lead Time Risk	Short and reliable lead time	Mostly reliable lead time	Moderate variability	Long or uncertain lead time	Very long or highly unpredictable
Tariff / Trade Cost	No tariff or import exposure	Low tariff exposure	Moderate trade exposure	High tariff uncertainty	High tariff / restriction risk
Contract Lead Time	Fast onboarding and flexible terms	Reasonable contract cycle	Average negotiation cycle	Slow or complex contracting	Very difficult access / long-term allocation only
Capacity Availability	Strong capacity available	Capacity mostly available	Capacity constrained but manageable	Major bottleneck risk	Severe bottleneck / allocation risk
Supplier Criticality	Important but not single point of failure	Important, with alternatives	Moderately critical	Highly critical	Essential and hard to operate without
Substitution Difficulty	Easy to replace	Alternatives exist	Some alternatives, switching cost	Few realistic substitutes	Near impossible to substitute short-term

A2. Full Scenario Results

Scenario	Region	Geo	Lead	Trade	Contract	Capacity	Criticality	Substitution	Score	Rank
Normal Conditions	Taiwan	2	3	4	3	2	5	1	2.80	3
Normal Conditions	South Korea	3	3	4	3	3	5	2	3.30	2
Normal Conditions	USA Domestic Hub	5	5	5	4	3	3	4	4.10	1
Geopolitical Shock	Taiwan	2	3	4	3	2	5	1	2.60	3
Geopolitical Shock	South Korea	3	3	4	3	3	5	2	3.20	2
Geopolitical Shock	USA Domestic Hub	5	5	5	4	3	3	4	4.35	1
Capacity Shortage	Taiwan	2	3	4	3	2	5	1	2.65	3
Capacity Shortage	South Korea	3	3	4	3	3	5	2	3.20	2
Capacity Shortage	USA Domestic Hub	5	5	5	4	3	3	4	3.95	1
Domestic Resilience Strategy	Taiwan	2	3	4	3	2	5	1	2.70	3
Domestic Resilience Strategy	South Korea	3	3	4	3	3	5	2	3.20	2
Domestic Resilience Strategy	USA Domestic Hub	5	5	5	4	3	3	4	4.30	1
Cost / Trade Pressure	Taiwan	2	3	4	3	2	5	1	2.90	3
Cost / Trade Pressure	South Korea	3	3	4	3	3	5	2	3.35	2
Cost / Trade Pressure	USA Domestic Hub	5	5	5	4	3	3	4	4.20	1

A3. Full Supplier Database With Source URLs

Supplier Group	Region	Company	Supply Chain Role	Product / Service	Direct or Upstream	Criticality Rationale	Source URL	Notes
Supplier A	Taiwan	TSMC	Foundry manufacturing	Advanced GPU / AI chip manufacturing	Direct strategic supplier	Advanced foundry capability is difficult to substitute.	https://blogs.nvidia.com/blog/tsmc-blackwell-manufacturing/	Core Taiwan supplier exposure.
Supplier A	Taiwan	TSMC CoWoS	Advanced packaging	CoWoS packaging for AI chips	Direct strategic supplier	Advanced packaging can become a capacity bottleneck.	https://www.tsmc.com/english/dedicatedFoundry/services/cowos	Packaging risk layer.
Supplier A	Taiwan	GlobalWafers	Silicon wafers	Silicon wafers	Upstream supplier	Wafer supply affects semiconductor output and lead times.	https://www.sas-globalwafers.com/	Upstream exposure.
Supplier A	Taiwan	Foxconn	System assembly	AI servers and electronics manufacturing	Downstream manufacturing partner	Assembly capacity affects delivery of final systems.	https://www.foxconn.com/en-us	Assembly layer.
Supplier A	Taiwan	Wistron	System assembly	Servers and electronics manufacturing	Downstream manufacturing partner	Potential assembly partner for AI infrastructure.	https://www.wistron.com/	Assembly layer.
Supplier B	South Korea	SK hynix	HBM memory	High-bandwidth memory	Direct strategic supplier	HBM is critical for AI accelerators.	https://www.skhynix.com/	Memory layer.
Supplier B	South Korea	Samsung Electronics	Memory / foundry	HBM, DRAM, foundry services	Direct or alternate supplier	Provides memory and semiconductor capacity.	https://semiconductor.samsung.com/	Alternate / diversification option.
Supplier B	South Korea	SK Siltron	Silicon wafers	Silicon wafers	Upstream supplier	Wafer supplier supporting semiconductor manufacturing.	https://www.sksiltron.com/	Upstream exposure.
Supplier C	USA	TSMC Arizona	Domestic foundry hub	U.S.-made advanced wafers / chips	Domestic hub	Reduces cross-border exposure for some production.	https://blogs.nvidia.com/blog/tsmc-blackwell-manufacturing/	U.S. resilience layer.
Supplier C	USA	Micron	Memory	DRAM / HBM roadmap exposure	Domestic / allied memory option	Potential U.S.-based memory diversification.	https://www.micron.com/	Memory diversification.
Supplier C	USA	Amkor	Advanced packaging	Semiconductor packaging and test	Domestic / regional hub	Packaging can reduce reliance on overseas bottlenecks.	https://amkor.com/	Packaging layer.
Supplier C	USA	Intel Foundry	Foundry manufacturing	Domestic foundry services	Domestic alternative	Potential long-term domestic foundry alternative.	https://www.intel.com/content/www/us/en/foundry/overview.html	Substitution option.
Supplier C	USA	DigiKey	Local distributor	Electronic components	Local seller	Useful for lower-level components and development needs.	https://www.digikey.com/	Local seller, not AI GPU supplier.
Supplier C	USA	Mouser	Local distributor	Electronic components	Local seller	Useful for small-volume component sourcing.	https://www.mouser.com/	Local seller, not AI GPU supplier.
Supplier C	USA	Arrow Electronics	Distributor / supply chain services	Electronic components and services	Local seller	Useful for sourcing support and supply chain services.	https://www.arrow.com/	Local seller.
Supplier C	USA	Avnet	Distributor / supply chain services	Electronic components and services	Local seller	Useful for electronics sourcing and supply chain services.	https://www.avnet.com/	Local seller.

Appendix B: Source List

The workbook source list is included below. These are the sources used in the model workbook. Before publishing this as a final professional research report, the source list should be checked and updated with the most current public data available.

Source	Type	URL	Used For	Notes
NVIDIA Blog: Blackwell wafer with TSMC Arizona	Official company source	https://blogs.nvidia.com/blog/tsmc-blackwell-manufacturing/	TSMC Arizona / U.S. domestic hub context	Use for current U.S. manufacturing context.
NVIDIA official site	Official company source	https://www.nvidia.com/	Case company background	General company context.
TSMC CoWoS page	Official company source	https://www.tsmc.com/english/dedicatedFoundry/services/cowos	Advanced packaging layer	Use for packaging explanation.
SK hynix official site	Official company source	https://www.skhynix.com/	HBM / memory supplier context	Use for South Korea supplier profile.
Samsung Semiconductor official site	Official company source	https://semiconductor.samsung.com/	South Korea semiconductor supplier profile	Use for diversification option.
GlobalWafers official site	Official company source	https://www.sas-globalwafers.com/	Taiwan wafer supplier context	Use for upstream wafer exposure.
SK Siltron official site	Official company source	https://www.sksiltron.com/	South Korea wafer supplier context	Use for upstream wafer exposure.
Micron official site	Official company source	https://www.micron.com/	U.S. memory supplier context	Use for domestic memory option.
Amkor official site	Official company source	https://amkor.com/	Advanced packaging / test	Use for domestic packaging option.
DigiKey official site	Distributor source	https://www.digikey.com/	Local seller comparison	Use for domestic distributor research.
Mouser official site	Distributor source	https://www.mouser.com/	Local seller comparison	Use for domestic distributor research.
Arrow official site	Distributor source	https://www.arrow.com/	Local seller comparison	Use for domestic distributor research.
Avnet official site	Distributor source	https://www.avnet.com/	Local seller comparison	Use for domestic distributor research.